GinCol Lab · Aula

Student worksheet

Physical Education · 1st year of Bachillerato · Your body, your lab · Class 2 of 4 · With Isidora

Your pulse, your body's first data point

Class video: https://youtu.be/2TtxZxspRXY

The code leads to the video.

Name: _____ Date: _____

Before you start

1. Without measuring anything yet: how many beats per minute do you think you have now, at rest? And what do you think is the fastest your heart can beat? Write it down: it's just for you.

2. **At the end of the class.** Go back to your two numbers. Were you close? What would you compare them with now: with a rule, or with people your age?

Your lab

What the video suggests, **if you feel well and want to do it**. Adapt it to you: if you have an injury or any health problem, don't do the effort; if you feel dizzy, your chest hurts, you're very out of breath or your knees hurt, stop.

Measurement	How long you count	Beats counted	×	Beats per minute
At rest, after sitting for a few minutes	30 s		× 2	
Right after 30 squats in 45 s	15 s		× 4	
One minute after finishing	15 s		× 4	

How much did your pulse go up with the effort? How much did it drop in the first minute? Did you win your bet (–10, –20 or –40)?

While you watch the video

1. **Everyone's formula.** Where did 220 – age come from? What did its authors write at the foot of their graph?

2. What is it that you feel in your wrist when you take your pulse? Why is counting the waves the same as counting the beats?

3. How do we know that the wave arrives a little after the heartbeat? How long does it take?

4. Why do you take your pulse with your index and middle fingers, and not with your thumb? Why do you never press both sides of your neck at once, and why do you press gently?

5. If you count for 15 seconds and multiply by 4, how much does one extra beat counted weigh? And if you count a whole minute?

6. **Rule 1.** What does the clinic's rule say about the resting pulse? How many healthy boys and girls aged 16 to 19 are below 60? What does it mean that it's "an agreement" and not "a law"?

7. Why are there gaps between the columns in the cloud of dots? What does that have to do with how you measured yourself?

8. What is the median resting pulse of girls aged 16 to 19? And of boys? Where do you fall?

9. Use the heart's calculation (blood per minute = blood per beat × beats per minute) to explain why a trained heart usually beats more slowly at rest.

10. After an effort, why do you count straight away and for only 15 seconds?

11. Why does your pulse go up when you move? And why does it drop so fast in the first minute after you stop? If you train for a few months, what usually happens to your pulse with the same effort? Why?

12. **Rule 2.** According to 220 – age, how much does your maximum drop each year? What has been measured in teenagers?

13. Why is Tanaka's formula, even though it's better, an extrapolation at your age?

14. Why do we say that a formula gives the average of lots of people and not your number?

Rule or data?

Mark **R** if it's a rule or a formula (an agreement or an estimate) and **D** if it's data measured in real people.

1. At rest, normal is 60 to 100 beats per minute.

2. The median resting pulse of boys aged 16 to 19, in the US survey, is 69.

3. Your maximum is 220 minus your age.

4. In 433 young athletes aged about 14, the measured maximum was, on average, 197.

5. At 16, Tanaka's formula gives about 197.

6. In the US survey, more than 1 in 10 boys aged 16 to 19 had less than 60 at rest.

Practice

1. **Two pulses.** A 16-year-old girl has 84 at rest and a boy, 60. Say where each one falls compared with people their age (the median: 77 for girls and 69 for boys), and whether either should be worried.

2. **Your maximum, with two formulas.** Work out your maximum with 220 − age and with Tanaka's formula (208 − 0.7 × age). How many beats separate them?

3. **Two ways of counting.** At rest, sitting down, count your pulse for 15 seconds and multiply by 4, and straight afterwards count it for a whole minute. Explain why you might not get the same result.

Do it yourself, if you want to

Three mornings in a row, after sitting quietly for a few minutes, like the people in the study, count your pulse for a whole minute. Work out the average and see where you fall compared with people your age.

	Day 1	Day 2	Day 3	Average
Resting pulse, sitting (1 minute)				

Your numbers are yours: nobody has to see them. If your resting pulse is often over 100, if it's very slow and you don't do much sport and you also feel dizzy, or if it feels very irregular, talk to your doctor about it.

Keep thinking

If a formula can be 17 beats off for you, what is it good for?

The data in the class come from the NHANES 1999–2008 survey (CDC, United States) and from PhysioNet (Vollmer et al., 2023), and from the studies cited in the teacher sheet.

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